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ENCLOSED REFLEX SIGHT FOR FIREARMS, ASSEMBLY, SYSTEM AND METHOD

Abstract

The disclosure is related to an enclosed reflex sight for one or more firearms. The enclosed reflex sight may be provided as part of an assembly including a mounting interface. The enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly, wherein the rearward bottom portion includes a lowermost surface of the enclosed reflex sight. The enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the mounting interface. When the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/372,472, filed on Sep. 25, 2023, which is entitled to the benefit of the filing date of the prior-filed U.S. Provisional Patent Application No. 63/412,351, filed on Sep. 30, 2022, the content of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] This disclosure relates generally in the field of optical sights for firearms.

2. Background Art

[0003] A reflex sight is a type of non-magnified (1x) electronic optical sight for firearms that is commercially available as an open reflex sight and as an enclosed reflex sight. An open reflex sight typically has a partly open housing with a single lens in an optical pathway of a user's eye, e.g., a shooter's eye, and an exposed light source on the housing. An enclosed reflex sight has a housing with multiple lenses in an optical pathway of a user's eye, and a light source located within the housing that assists in protecting the light source against environmental influences such as dirt and moisture.

[0004] As understood by the skilled artisan, a housing of an enclosed reflex sight provides a field of view including a projected reticle to assist in aiming a corresponding firearm. Some enclosed reflex sights include windage and elevation adjustment mechanisms within the housing below the field of view. Other enclosed reflex sights house a light source in a pivot tube that allows the windage and elevation adjustment mechanisms to be located at the top of the housing or on the side in line with the field of view. In both configurations, the windage and elevation adjustment mechanisms are operationally configured to assist a user in matching a projectile impact for a firearm to an aiming point of the closed reflex sight.

[0005] In typical operation, an enclosed reflex sight is secured to an upper mounting surface of a firearm via a mounting interface at a position forward of a protruding rear sight, i.e., a rear iron sight, of the firearm. In one known implementation, an upper mounting surface may include an upper surface 555 of a firearm 550 whereby a mounting interface 525 is secured directly to the upper surface 555 via one or more threaded fasteners 556 forward of a rear sight 530 (see the mounting interface 525 releasably secured to an upper surface 555 of a pistol slide 575 in FIG. 1). As shown in FIG. 2, in another known implementation an upper mounting surface 557 of a particular size and shape may be formed along an upper surface 555 of a firearm 550 for receiving a mounting interface 525 (see FIGS. 2 and 3). In this implementation, the upper mounting surface 557 commonly includes a bottom surface 570, a forward or front wall surface 571 and a rearward or back wall surface 572 located forward of a rear sight 530 operationally configured as a mounting surface envelope for receiving a mounting interface 525 for securing an enclosed reflex sight at a position forward of a rear sight 530 as shown in FIG. 3.

[0006] In each of the implementations described above, the entire enclosed reflex sight 500 is located at a point between a rear end surface 538 and a front end surface 539 of the firearm 550 on top of the mounting interface 525 forward the rear sight 530 whereby the mounting interface 525 spaces the enclosed reflex sight apart from the upper mounting surface of the firearm. Because the elevation adjustment mechanism of a known enclosed reflex sight 500 as shown in FIG. 3 operates

in a vertical direction, the housing must include enough space to accommodate the elevation adjustment mechanism. Unfortunately, an enlarged housing raises the location of an optical center 505 of a clear aperture or usable field of view of the enclosed reflex sight 500 in relation to both the upper mounting surface of the firearm 550 (see the bottom surface 570 of the upper mounting surface 557 in FIG. 3) and the top of the original equipment (“OE”) standard height iron sights for a particular firearm thereby preventing the rear and front sights from being used for purposes of co-witnessing with the enclosed reflex sight 500 (see the sighting line 540 of the standard height rear sight 530 and front sight 531 and distance 541 in FIG. 3; also see the no-witness position between the standard height iron sights 530, 531 and the reticle 535 of the enclosed reflex sight 500 in FIG. 4). To achieve co-witnessing, the standard height iron sights 530, 531 must be replaced with taller aftermarket rear and front sights 532 and 533 as shown in FIG. 5.

[0007] In consideration of the above described enclosed reflex sight 500 design, another enclosed reflex sight 590 known in the art houses electronics and adjustment mechanisms in a rear section 591 that extends beyond the upper surface 555 of a firearm 550 behind a rear end surface 538 of a pistol slide 575 as shown in the simplified illustrations of FIGS. 6 and 7. However, as shown, an enclosed reflex sight 590 of this embodiment may require adjustment in design to accommodate pistol slide 575 geometry of a particular firearm 550. In addition, the corresponding firearm 550 does not employ a rear sight.

[0008] An enclosed reflex sight that overcomes the above shortcomings while also being operable with a plurality of firearm mounting footprints is desired.

SUMMARY OF THE DISCLOSURE

[0009] The present disclosure provides an assembly comprising an enclosed reflex sight and a mounting interface; wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight; wherein the enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the mounting interface; and wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.

[0010] The present disclosure also provides an assembly comprising an enclosed reflex sight and a mounting interface; wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the enclosed reflex sight is located at a position closer to the upper mounting surface of the firearm than at least part of the mounting interface and the assembly is located forward of a rear end surface of the firearm.

[0011] The present disclosure also provides an assembly comprising an enclosed reflex sight and a plurality of mounting interfaces; wherein each of the plurality of mounting interfaces is configured to secure the assembly to one or more of a plurality of firearms comprising a plurality of upper mounting surface footprints; wherein when the assembly is secured to an upper mounting surface of a first firearm of the plurality of firearms then the assembly is located on top of the first firearm forward of a rear end surface of the first firearm; and wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0012] The present disclosure also provides a system comprising (1) at least one firearm and (2) an assembly comprising (a) an enclosed reflex sight and (b) at least one mounting interface; wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight; wherein the enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the at least one mounting interface; and wherein when the

assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.

[0013] The present disclosure also provides a system comprising (1) at least one firearm comprising an upper mounting surface and (2) an assembly securable to the upper mounting surface comprising (a) an enclosed reflex sight and (b) at least one firearm mounting interface; wherein the assembly includes a maximum height and wherein the enclosed reflex sight includes a maximum height the same or substantially similar as the maximum height of the assembly; and wherein the assembly is located forward of a rear end surface of the firearm.

[0014] The present disclosure also provides a system comprising (1) a plurality of firearms providing at least two different upper mounting surfaces, (2) an enclosed reflex sight and (3) a plurality of mounting interfaces, each mounting interface of the plurality of mounting interfaces being configured to secure the enclosed reflex sight to at least one upper mounting surface of the at least two different upper mounting surfaces; wherein when the enclosed reflex sight is secured to the at least one upper mounting surface of a first firearm of the plurality of firearms via a first mounting interface of the plurality of mounting interfaces then the enclosed reflex sight is located on top of the first firearm forward of a rear surface of the first firearm wherein a lowermost surface of the enclosed reflex sight is located closer to the at least one upper mounting surface than at least part of the first mounting interface.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEW OF THE DRAWING

[0015] FIG. 1 is a perspective view of a prior art mounting interface 525 secured to an upper surface 555 of a pistol slide 575 of a firearm 550.

[0016] FIG. 2 is a perspective view of a pistol slide 575 including a mounting surface 557.

[0017] FIG. 3 is a side view illustration of a prior art enclosed reflex sight 500 secured to a slide component of a semi-automatic pistol type firearm 550 via a prior art mounting interface 525.

[0018] FIG. 4 is a rear view of the enclosed reflex sight 500 of FIG. 1 in a no-witness position in relation to rear and front sights 530, 531 of the firearm 550.

[0019] FIG. 5 is a rear view of the enclosed reflex sight of FIG. 1 comprising taller prior art rear and front iron sights 532, 533 illustrating a lower $\frac{1}{2}$ co-witness of the reticle 535 with the rear and front iron sights 532, 533.

[0020] FIG. 6 is a side view of an embodiment of a prior art enclosed reflex sight 590 for a firearm 550 including part of the enclosed reflex sight 590 located beyond or off from an upper mounting surface of a firearm 550 behind a pistol slide 575 when the firearm 550 is oriented in a firing position.

[0021] FIG. 7 is a side view of an embodiment of a prior art enclosed reflex sight 590 for a firearm 550 including part of the enclosed reflex sight 590 located beyond or off from an upper mounting surface of a firearm 550 behind a pistol slide 575 when the firearm 550 is oriented in a firing position.

[0022] FIG. 8 is a front top perspective view of an embodiment of an assembly of the present disclosure.

[0023] FIG. 9 is a front bottom perspective view of the assembly of FIG. 8.

[0024] FIG. 10 is a left side elevation view of the assembly of FIG. 8.

[0025] FIG. 11 is an exploded view of the assembly of FIG. 8.

[0026] FIG. 12 is an exploded view of part of the assembly of FIG. 8.

[0027] FIG. 13 is a cutaway view of the housing of the enclosed reflex sight of the assembly of FIG. 8.

[0028] FIG. **14** is a cutaway view of the housing of the enclosed reflex sight of the assembly of FIG. **8**.

[0029] FIG. **15** is a bottom perspective view of part of the enclosed reflex sight of the assembly of FIG. **8**.

[0030] FIG. **16** is a bottom view of part of the enclosed reflex sight of the assembly of FIG. **8**,

[0031] FIG. **17** is a bottom perspective view of part of the enclosed reflex sight of the assembly of FIG. **8**.

[0032] FIG. **18** is a bottom perspective view of part of the enclosed reflex sight of the assembly of FIG. **8**.

[0033] FIG. **19** is a bottom view of part of the enclosed reflex sight of the assembly of FIG. **8**,

[0034] FIG. **20** is a bottom perspective view of part of the enclosed reflex sight of the assembly of FIG. **8**.

[0035] FIG. **21** is a bottom perspective view of the enclosed reflex sight of the assembly of FIG. **8**.

[0036] FIG. **22** is a left side elevation view of the enclosed reflex sight of the assembly of FIG. **8**.

[0037] FIG. **23** is a right side elevation view of the enclosed reflex sight of the assembly of FIG. **8**.

[0038] FIG. **24** is a left side elevation view of the assembly of FIG. **8**.

[0039] FIG. **25** is a right side elevation view of the assembly of FIG. **8**.

[0040] FIG. **26** is an exploded view of an adjustment assembly and bottom plate of the enclosed reflex sight of the assembly of FIG. **8**.

[0041] FIG. **27** is a perspective sectional view of part of the adjustment assembly of FIG. **26**.

[0042] FIG. **28** is a perspective view of the adjustment assembly of FIG. **26**,

[0043] FIG. **29** is a perspective view of the adjustment assembly of FIG. **26**.

[0044] FIG. **30** is a perspective view of the bottom plate of FIG. **26**.

[0045] FIG. **31** is a bottom view of part of the enclosed reflex sight of the assembly of FIG. **8**,

[0046] FIG. **32** is a perspective view of the mounting interface of the assembly of FIG. **8**.

[0047] FIG. **33** is a top view of the mounting interface of FIG. **32**.

[0048] FIG. **34** is a bottom view of the mounting interface of FIG. **32**.

[0049] FIG. **35** is a perspective view of an embodiment of a mounting interface for assembly with the enclosed reflex sight of FIG. **8**.

[0050] FIG. **36** is a top view of the mounting interface of FIG. **35**.

[0051] FIG. **37** is a bottom view of the mounting interface of FIG. **35**.

[0052] FIG. **38** is a perspective view of an embodiment of a mounting interface for assembly with the enclosed reflex sight of FIG. **8**.

[0053] FIG. **39** is a top view of the mounting interface of FIG. **38**.

[0054] FIG. **40** is a bottom view of the mounting interface of FIG. **38**.

[0055] FIG. **41** is a rear elevation view of an embodiment of an enclosed reflex sight of this disclosure.

[0056] FIG. **42** is a front perspective view of the enclosed reflex sight of FIG. **41**.

[0057] FIG. **43** is a top view of the enclosed reflex sight of FIG. **41**.

[0058] FIG. **44** is a right side elevation view of the enclosed reflex sight of FIG. **41**.

[0059] FIG. **45** is a bottom view of the enclosed reflex sight of FIG. **41**.

[0060] FIG. **46** is a left side elevation view of the enclosed reflex sight of FIG. **41**.

[0061] FIG. **47** is a front elevation view of the enclosed reflex sight of FIG. **41**.

[0062] FIG. **48** is a bottom perspective view of the enclosed reflex sight of FIG. **41**.

[0063] FIG. **49** is a rear elevation view of the assembly of FIG. **8**.

[0064] FIG. **50** is a front top perspective view of the assembly of FIG. **49**.

[0065] FIG. **51** is a top view of the assembly of FIG. **49**.

[0066] FIG. **52** is a right side elevation view of the assembly of FIG. **49**.

[0067] FIG. **53** is a bottom view of the assembly of FIG. **49**.

[0068] FIG. **54** is a left side elevation view of the assembly of FIG. **49**.

[0069] FIG. 55 is a front elevation view of the assembly of FIG. 49.

[0070] FIG. 56 is a front bottom perspective view of the assembly of FIG. 49.

[0071] FIG. 57 is a rear perspective view of an embodiment of a semi-automatic pistol slide.

[0072] FIG. 58 is a left side view of the assembly of FIG. 8 secured to the semi-automatic pistol slide of FIG. 57.

[0073] FIG. 59 is a rear left side perspective view of the assembly of FIG. 8 secured to the semi-automatic pistol slide of FIG. 57.

[0074] FIG. 60 is a rear right side perspective view of the assembly of FIG. 8 secured to the semi-automatic pistol slide of FIG. 57.

[0075] FIG. 61 is a rear view of a side by side comparison including the assembly of FIG. 8 secured to the semi-automatic pistol slide of FIG. 57 on the left and the prior art enclosed reflex sight 500 of FIG. 3 secured to the semi-automatic pistol slide of FIG. 57 on the right.

[0076] FIG. 62 is a rear elevation view of the enclosed reflex sight of the assembly of FIG. 8.

[0077] FIG. 63 is a right side elevation view of the enclosed reflex sight of the assembly of FIG. 8.

DEFINITIONS USED IN THE DISCLOSURE

[0078] The term “at least one”, “one or more”, and “one or a plurality” mean one thing or more than one thing with no limit on the exact number; these three terms may be used interchangeably within this disclosure. For example, at least one device means one or more devices or one device and a plurality of devices.

[0079] The term “about” means that a value of a given quantity is within $\pm 20\%$ of the stated value. In other embodiments, the value is within $\pm 15\%$ of the stated value. In other embodiments, the value is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 7.5\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value.

[0080] The term “substantially” or “essentially” means that a value of a given quantity is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 7.5\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value. In other embodiments, the value is within $\pm 0.5\%$ of the stated value. In other embodiments, the value is within $\pm 0.1\%$ of the stated value.

[0081] The term “and/of” includes any and all combinations of one or more of the associated listed items.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0082] For the purposes of promoting an understanding of the principles of the disclosure, reference is now made to the embodiments illustrated in the drawings and particular language will be used to describe the same. It is understood that no limitation of the scope of the claimed subject matter is intended by way of the disclosure. As understood by one skilled in the art to which the present disclosure relates, various changes and modifications of the principles as described and illustrated are herein contemplated.

[0083] It is to be understood that the present disclosure is not limited to particular embodiments. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. Herein, a “firearm” may include, but is not limited to a pistol, a semiautomatic firearm, e.g., a semiautomatic rifle, a bolt action firearm, e.g., a bolt action rifle, a shotgun, a revolver, a shoulder fired bazooka, a shoulder fired rocket launcher, an air rifle, and a paintball gun. As understood by the skilled artisan, a particular firearm may be provided in different barrel lengths. Non-limiting examples of pistols are provided in U.S. Pat. No. 4,539,889, titled “Automatic Pistol with Counteracting Spring Control Mechanism,” issued on Sep. 10, 1985; and U.S. Pat. No. D918,328 S, titled “Handgun,” issued on May 4, 2021, each of which is herein incorporated by reference in its entirety. Non-limiting examples of semiautomatic rifles

are provided in U.S. Pat. No. 9,777,975 B2, titled “Semiautomatic Firearm,” issued on Oct. 3, 2017; and U.S. Pat. No. 7,775,150 B2, titled “Law Enforcement Carbine with One Piece Receiver,” issued on Aug. 17, 2010, each of which is herein incorporated by reference in its entirety.

[0084] In this disclosure, reference to an “optical sight” includes an optical sight for a firearm or a “firearm optical sight.” Reference to one or more “commercially available” enclosed reflex sights and/or other optical sights includes those optical sights commercially available as of the time of this disclosure. Herein, “OE” stands for original equipment and “OEM” stands for original equipment manufacturer. In this disclosure, the terms “button cell battery,” “button cell,” “button style battery,” “coin cell battery,” “coin cell” and “coin style battery” may be used interchangeably.

[0085] Herein, an upper mounting surface of a firearm 550 for use with an enclosed reflex sight of this disclosure is a mounting surface located along an upper surface 555 of a firearm 550 between a rear sight 530 and a front end 539 of the firearm 550 for locating the entire enclosed reflex sight on top of the firearm 550 forward of the rear sight 530. In an embodiment of a firearm 550 devoid of a rear sight 530, an upper mounting surface of a firearm 550 for an enclosed reflex sight of this disclosure is a mounting surface located along an upper surface 555 of a firearm 550 from or about a rear end of the upper surface 555 of the firearm 550 forward to or about the front end of the upper surface 555 of the firearm 550 for locating the entire enclosed reflex sight on top of the firearm 550. For purposes of this disclosure, an upper mounting surface 557 as shown in FIGS. 2 and 3 may be referred to herein as an “upper mounting surface envelope” whereby an enclosed reflex sight of this disclosure alone or as part of an assembly including a corresponding mounting interface operates as an upper mounting surface inset for the upper mounting surface envelope locating the entire enclosed reflex sight or the assembly on top of a firearm 550 forward of a rear end surface 538 of a firearm 550 and also forward of a rear sight 530 of a firearm 550 when a rear sight 530 is included as part of a firearm 550.

[0086] In this disclosure, an “illuminated reticle” refers to a reticle of an optical sight that may be illuminated via electrical components including a power source and one or more electrically powered and electrically controlled light sources of the optical sight. Exemplary light sources may include, but are not limited to one or more incandescent bulbs, one or more fluorescent bulbs, one or more light emitting diodes (“LEDs”), one or more resonant cavity light-emitting diodes (“RCLEDs”), one or more diode lasers, one or more organic LEDs, one or more vertical cavity surface emitting laser diodes (“VCSELs”), and combinations thereof. In one non-limiting embodiment, a reticle may be illuminated via a light emitting diode (“LED”) or an LED array and/or (2) a resonant cavity light-emitting diode (“RCLED”) or an RCLED array. In another embodiment, a reticle may be illuminated via a diode laser, e.g., a reticle of a holographic optical sight. In another embodiment, a reticle may be illuminated via a liquid crystal display (“LCD”). In another embodiment, a reticle may be illuminated via an organic light-emitting diode (“OLED”) display. In another embodiment, a reticle may be illuminated via a vertical cavity surface emitting laser diode (“VCSEL”). An LED of this disclosure may include a single color LED or a multi-color LED, e.g., a dual color LED, a tri-color LED. An RCLED of this disclosure may include a single color RCLED or a multi-color RCLED. An LED array of this disclosure may include one or more single color LEDs, one or more multi-color LEDs, and combinations thereof. An RCLED array of this disclosure may include one or more single color RCLEDs, one or more dual color RCLEDs, one or more tri-color RCLEDs, and combinations thereof. Exemplary LED and/or RCLED colors may include, but are not limited to red, green, yellow, blue, cyan, orange, and combinations thereof. One or more other commercially available LED and/or RCLED colors may also be employed as desired.

[0087] Herein, the phrase “field of view” (“FOV”) refers to the visible or observable area through an optical element or lens or lenses of an optical sight for an operator of the optical sight, e.g., a shooter, at a particular distance between the operator's eye(s) and the optical element or lens of the optical sight. Herein, “clear aperture” refers to the smallest diameter, width, or area inside an

enclosed reflex sight that creates the optical viewing area of the enclosed reflex sight.

[0088] The phrase “co-witnessing” and like terms refers to the interaction between a reticle of an optical sight and iron sights of a corresponding firearm. As understood by the skilled artisan, an optical sight comprises a co-witness when the reticle of the optical sight is aligned with the iron sights of the corresponding firearm. As used in this disclosure, the term “reticle” in relation to firearms refers to lines and/or other markings or indicia found on an optical element of an optical sight. One exemplary reticle for use with a reflex sight including an enclosed reflex sight includes a luminous marking such as a luminous dot, e.g., a red dot or other colored dot. Regarding an enclosed reflex sight of this disclosure, the phrase “operable eye distance” refers to one or more distances between a shooter's eye and a front lens of an enclosed reflex sight effective for suitable operation of a reticle projected onto the front lens of the enclosed reflex sight. As such, the field of view of the lenses of an enclosed reflex sight at an operable eye distance may be referred to herein as an “operable field of view” or “clear aperture.” The operable eye distance and the operable field of view may vary depending on the optical sight and firearm combination employed. For example, an operable eye distance for an optical sight and pistol combination may be greater than an operable eye distance for an optical sight and rifle combination.

[0089] In an embodiment, the disclosure is related to an enclosed reflex sight system operable for co-witnessing with protruding iron sights of one or more firearms.

[0090] In another embodiment, the disclosure is related to an enclosed reflex sight assembly operable for co-witnessing with protruding iron sights of one or more firearms.

[0091] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface having a length less than the length of the enclosed reflex sight whereby the windage and elevation adjustment mechanisms of the enclosed reflex sight are located behind the firearm mounting interface closer to a corresponding firearm when the assembly is mounted to a firearm compared to known commercially available enclosed reflex sights whereby the present enclosed reflex sight is operable for co-witnessing with standard height iron sights.

[0092] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface operationally configured to provide a shorter and less bulky design of an enclosed reflex sight compared to known commercially available enclosed reflex sights when mounted to a firearm such as a semi-automatic pistol.

[0093] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface wherein the firearm mounting interface is located forward of a rearmost part of the enclosed reflex sight when the enclosed reflex sight is assembled to the firearm mounting interface.

[0094] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface wherein the maximum height of the enclosed reflex sight is the same or substantially similar as the maximum height of the assembly of the enclosed reflex sight and the firearm mounting interface. Said another way, the distance between the optical center of a clear aperture of the enclosed reflex sight and the bottommost surface of the enclosed reflex sight is the same or substantially similar as the distance between the optical center of the clear aperture of the enclosed reflex sight and the bottommost surface of the firearm mounting interface when the enclosed reflex sight is assembled to the firearm mounting interface. Although an assembly of this disclosure may be built to scale, in an embodiment of the assembly operationally configured for use with a semi-automatic pistol type firearm **550**, any difference in the distances of the bottommost surface of the enclosed reflex sight and the bottommost surface of the firearm mounting interface is not distinguishable to the unaided eye including when the assembly is mounted to a firearm, e.g., see FIGS. **58-60**. As stated above, although an assembly of this disclosure may be built to scale, in an embodiment of an assembly operationally configured for use with a semi-automatic pistol type firearm **550** any measurable difference in distance from the

optical center of a clear aperture of the enclosed reflex sight to the bottommost surface of the firearm mounting interface and the bottommost surface of the enclosed reflex sight is a distance of or about 5.0 mm or less.

[0095] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a mounting interface for securing the enclosed reflex sight to at least a first firearm. In this embodiment, the enclosed reflex sight comprises windage and elevation adjustment mechanisms in a rearward bottom portion of the enclosed reflex sight wherein when the enclosed reflex sight is secured to the mounting interface then the rearward bottom portion of the enclosed reflex sight is located behind the mounting interface and wherein the windage and elevation adjustment mechanisms operate in the horizontal direction. The configuration of the windage and elevation adjustment mechanisms operating in the horizontal direction allows for an overall height reduction of the enclosed reflex sight compared to the height of prior art enclosed reflex sights.

[0096] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface. In another embodiment, when the enclosed reflex sight is secured to the firearm mounting interface and mounted to a firearm oriented in a firing position, a lowermost bottom surface of the rearward part of the enclosed reflex sight is located at an altitude or elevation lower than at least part of the firearm mounting interface. In another embodiment, when the enclosed reflex sight is secured to the firearm mounting interface and mounted to a firearm oriented in a firing position, the lowermost bottom surface of the forward part of the enclosed reflex sight is located at an altitude or elevation lower than at least part of the firearm mounting interface.

[0097] In another embodiment, the disclosure is related to an enclosed reflex sight for a firearm and a mounting interface for the enclosed reflex sight, wherein the enclosed reflex sight includes a bottom surface having a first bottom surface and a second bottom surface. In this embodiment, when the enclosed reflex sight and the mounting interface are mounted to a firearm that is oriented in a firing position then the first bottom surface is located at one or more first altitudes or elevations and the second bottom surface is located at one or more second altitudes or elevations lower than the one or more first altitudes or elevations, the first bottom surface comprising an abutment surface for the mounting interface. In this embodiment, the mounting interface includes a bottom surface wherein when the enclosed reflex sight is secured to the mounting interface then the bottom surface of the mounting interface is in planar alignment or substantially planar alignment with the second bottom surface of the enclosed reflex sight.

[0098] In another embodiment, the disclosure is related to an enclosed reflex sight for a firearm and a mounting interface for the enclosed reflex sight, wherein the enclosed reflex sight includes a bottom surface having a front section bottom surface and a rear section bottom surface. In this embodiment, when the enclosed reflex sight and the mounting interface are mounted to a firearm that is oriented in a firing position then the front section bottom surface is located at a first altitude or elevation and the rear section bottom surface is located at a second altitude or elevation lower than the first altitude or elevation, the front section bottom surface comprising an abutment surface for the mounting interface. In this embodiment, the mounting interface includes a bottom surface wherein when the enclosed reflex sight is secured to the mounting interface then the bottom surface of the mounting interface is in planar alignment or substantially planar alignment with the rear section bottom surface of the enclosed reflex sight.

[0099] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface with a length less than the length of the enclosed reflex sight whereby the windage and elevation adjustment mechanisms of the enclosed reflex sight are located behind the firearm mounting interface and whereby each of the adjustment mechanisms operate in a horizontal direction effective to minimize the size of the present enclosed reflex sight compared to known commercially available enclosed reflex sights comprising one or more adjustment mechanisms operating in a vertical direction.

[0100] In another embodiment, the disclosure is related to a system comprising an enclosed reflex sight and a plurality of firearm mounting interfaces configured for use with a plurality of firearms, e.g., a plurality of semi-automatic pistols, whereby the enclosed reflex sight may be used with a plurality of semi-automatic pistols. In other words, the system is operationally configured so that the enclosed reflex sight may be used with one or more optical sight mounting footprints on firearms as known in the firearm industry at the time of this disclosure.

[0101] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a firearm mounting interface whereby, when mounted to a firearm such as a semi-automatic pistol, and depending on the semi-automatic pistol used, the optical center of a clear aperture of the enclosed reflex sight is located above the firearm a distance effective for the enclosed reflex sight to be used for co-witnessing with protruding iron sights.

[0102] In another embodiment, the disclosure is related to an assembly comprising an enclosed reflex sight and a chassis type firearm mounting interface having a length less than the length of the enclosed reflex sight whereby the windage and elevation adjustment mechanisms of the enclosed reflex sight are located behind the chassis type firearm mounting interface and whereby the enclosed reflex sight is releasably fastenable to the chassis type firearm mounting interface and corresponding firearm via one or more fasteners at one or more locations forward of the location of the windage and elevation adjustment mechanisms.

[0103] In another embodiment, the disclosure is related to a system including (1) an enclosed reflex sight for a firearm, wherein the enclosed reflex sight includes a forward bottom surface and a rearward bottom surface, wherein the rearward bottom surface is a lowermost outer surface of the enclosed reflex sight, and (2) a firearm comprising an upper surface machined or milled to include a mounting interface for the forward bottom surface of the enclosed reflex sight.

[0104] FIGS. **8-31** illustrate an embodiment of an assembly **10** of the present disclosure operationally configured for use with firearms **550** including, but not limited to semi-automatic pistols. In this embodiment, the assembly **10**, also referred to herein as the “enclosed reflex sight assembly **10**,” includes an enclosed reflex sight **11** and at least one mounting interface **14** for mounting or otherwise releasably securing the enclosed reflex sight to one or more firearms **550**. In particular, the enclosed reflex sight **11** comprises a main housing **12** (or “housing **12**”) releasably securable to the mounting interface **14** via one or more fasteners **15** wherein the mounting interface **14** is releasably securable to one or more firearms **550** via one or more firearm rail mounts known in the art of firearms, e.g., one or more dovetail mounts, one or more fasteners, one or more intermediary mounting interfaces, and combinations thereof.

[0105] With particular reference to FIGS. **11** and **12**, the enclosed reflex sight **11** comprises a first non-magnifying optical element (hereafter “front lens **16**”) disposed within a forward part of the housing **12** and a second non-magnifying optical element (hereafter “rear lens **18**”) disposed within a rearward part of the housing **12**. Suitably, the front lens **16** and the rear lens **18** are sealed within the housing **12** providing an optical space **34** there between including an optical path **199** there through (see FIG. **10**). Although the one or more operating components may vary according to a desired operation of a particular enclosed reflex sight **11**, in one embodiment the one or more operating components include at least control circuitry, an illumination system for a projected reticle, a power input system, and windage and elevation adjustment mechanisms. In another embodiment, an enclosed reflex sight **11** may also include one or more input/output ports, buttons, dials, switches, and combinations thereof.

[0106] With particular reference to FIGS. **11** and **12**, the control circuitry may include one or more primary or main printed circuit boards **20** (“PCB **20**”) comprising one or more design specifications including a patterned arrangement of printed circuitry and solid state electronic components mounted to the surface of the PCB **20**. As discussed further below, when supplied power via the power input system, a PCB **20** of an enclosed reflex sight **11** of this disclosure may be operationally configured to control or regulate the output power, the output voltage and the rate

of electric current of the enclosed reflex sight **11**, or pulse width modulation. One suitable PCB **20** may include an integrated circuit such as a microcontroller unit (“MCU”), peripheral hardware including one or more input devices and one or more other PCB components for desired operation of an enclosed reflex sight **11**. A MCU of this disclosure may include, but is not limited to a central processing unit (“CPU”), random-access memory (“RAM”), flash memory, a serial bus interface, input/output ports (“f/O Ports”), a special function register (“SFR”) for controlling peripheral circuitry, read-only memory (“ROM”), one or more voltage regulator circuits, one or more resistors, and combinations thereof. One or more input devices to implement one or more peripheral functions may include, but are not limited to one or more motion sensors, one or more wireless control circuits, one or more power regulation circuits, and combinations thereof.

[0107] In one embodiment, the PCB **20** may be secured within the housing **12** via one or more retaining clips, one or more threaded fasteners, one or more adhesives, and combinations thereof as known in the art of optical sights. In the embodiment as shown in FIG. **11**, the PCB **20** includes one or more through holes **21** for receiving one or more threaded fasteners **23** there through for releasably securing the PCB **20** to threaded holes **52** (see FIG. **15**) of the housing **12**. To protect the PCB **20** and other operating components within the housing **12**, the enclosed reflex sight **11** comprises a cover or plate member **22** secured to a corresponding mating surface of the housing **12** corresponding in size and shape as the plate member **22** via one or more adhesives (see FIG. **20**). In one embodiment, one or more adhesives are disposed along the perimeter of an inner surface **24** of the plate member **22** for adhering the perimeter of the inner surface **24** to the mating surface **68** providing a continuous seal for insulating the PCB **20** and other components in the housing **12** from environmental influences such as dirt and moisture.

[0108] An illumination system of this embodiment includes at least an emitter assembly **25** in electrical communication with the PCB **20** and securable to an emitter assembly carrier **29**, illumination circuitry including an illumination PCB **32** in electrical communication with the emitter assembly **25**, and one or more user controls **36** in mechanical communication with the illumination PCB **32**. In this embodiment, the emitter assembly **25** includes an emitter PCB **26** and one or more illumination sources **27** in electrical communication with the emitter PCB **26**. As common in electric circuitry, the emitter assembly **25** is electrically communicated with the PCB **20** via one or more electrical connectors such as one or more wire leads **28** providing power and logic signals to the emitter PCB **26**.

[0109] In one embodiment, the one or more illumination sources **27** may include one or more LEDs, one or more RCLEDs, and combinations thereof. For purpose of discussion, one non-limiting emitter assembly **25** of this disclosure may comprise one or more RCLEDs **27** operationally configured to generate a desired reticle onto an inner surface of the front lens **16**. As understood by persons of ordinary skill in the art of RCLEDs, each individual RCLED of the one or more RCLEDs **27** has a positive lead at one end of the RCLED and a negative lead at an opposite end of the RCLED providing electrical communication of the one or more RCLEDs **27** with the emitter PCB **26**.

[0110] As understood by persons of ordinary skill in the art of enclosed reflex sights, the emitter assembly **25** is located within the housing **12** at a location operable for projecting an operable reticle onto an inner surface **19** of the front lens **16**. As such, the one or more RCLEDs **27** of the emitter assembly **25** may be provided in a size and shape effective to project a reticle of a desired angular measurement onto the inner surface **19** of the front lens **16** according to (1) the dimensions of the front lens **16** and (2) the distance between the front lens **16** and a projection point of the one or more RCLEDs **27** to provide an observable reticle at a predetermined operable eye distance for the enclosed reflex sight **11**. In another embodiment, the one or more RCLEDs **27** of the emitter assembly **25** may be provided in a size and shape effective to project a reticle of a desired angular measurement onto the inner surface **19** of the front lens **16** according to (1) the dimensions of the front lens **16** and (2) the distance between the one or more RCLEDs **27** and a dichroic coating

disposed between a first lens element and a second lens element of the front lens **16**. In an embodiment of the emitter assembly **25** comprising one or more LEDs and an accompanying light blocking or masking element, the one or more openings of the light blocking plate may be provided in a size and shape effective to project a reticle of a desired angular measurement onto the inner surface **19** of the front lens **16** according to (1) the dimensions of the front lens **16** and (2) the distance between the front lens **16** and the one or more openings of the light blocking plate.

[0111] Suitably, the one or more user controls **36** are disposed along the housing **12** for manual operation allowing a user to manually control the illumination intensity or brightness of the one or more illumination sources **27** and a corresponding reticle of the enclosed reflex sight **11**. In one embodiment, the illumination PCB **32** may include (1) a positive setting switch **50** (or “first push switch **50**”) operationally configured to cycle the one or more illumination sources **27** through a series of increasing reticle illumination brightness settings and (2) a negative setting switch **51** (or “second push switch **51**”) operationally configured to cycle the one or more illumination sources through a series of decreasing reticle illumination brightness settings.

[0112] Referring to FIG. **11**, in one embodiment the one or more user controls **36** may be provided as a touch pad member **36** having a first designated push button **54** (or “first designated push area **54**”) operationally configured to be manually pressed inward to actuate the first push switch **50** and a second designated push button **55** (or “second designated push area **55**”) operationally configured to be manually pressed inward to actuate the second push switch **51**. In one embodiment, a touch pad member **36** may be constructed of one or more flexible materials. In another embodiment, at least the first designated push button **54** and the second designated push button **55** of the touch pad member **36** may be constructed of one or more flexible materials. Suitable flexible materials include one or more polymeric materials, one or more metals, and combinations thereof.

[0113] As shown in FIG. **11**, in one embodiment the first designated push button **54** and the second designated push button **55** of the touch pad member **36** may each include one or more numbers, one or more symbols, one or more color indicators, one or more words indicating the functions of each of the push buttons **54** and **55**. In this embodiment, the outer surface of the first designated push button **54** comprises a plus sign symbol (“+”) indicating the increasing reticle illumination brightness settings of the first push switch **50** and the outer surface of the second designated push button **55** comprises a minus sign symbol (“−”) indicating the decreasing reticle illumination brightness settings of the second push switch **51**. In another embodiment, push button **54** may include an upward pointing arrow or other indicator and push button **55** may include a downward pointing arrow or other indicator. In operation, a user may press the first designated push button **54** inward to actuate the first push switch **50** in order to cycle the one or more illumination sources **27** through a plurality of increasing illumination brightness settings and press the second designated push button **55** inward to actuate the second push switch **51** through a plurality of decreasing illumination brightness settings. In one embodiment, the first push switch **50** and the second push switch **51** may also include ON and OFF settings realized when cycling through the reticle illumination brightness settings. In one embodiment, at least one push button **54** and/or **55** may be operationally configured to cycle through a plurality of reticle configurations displaying different indicia.

[0114] With further reference to FIG. **11**, the first side **13** of the housing **12** includes a cavity type mating surface **30** operationally configured to receive the illumination PCB **32** and the touch pad member **36** therein. As shown, the mating surface **30** includes a planar floor **33** operationally configured as a contact surface for the illumination PCB **32** and an opening **35** in the floor **33** providing for electrical communication between the illumination PCB **32** and the emitter assembly **25** and the PCB **20** located within the housing **12** via electrical connectors or solder joints disposed through the opening **35**. Exemplary electrical connectors may include PCB wire-to-board connectors including electrical wiring and/or ribbon cable assemblies as known in the art of printed circuit boards. In another embodiment, the mating surface **30** may comprise a non-planar floor. In

another embodiment, the mating surface **30** may include a plurality of openings there through.

[0115] In one embodiment, the illumination PCB **32** may be secured to the floor **33** of the mating surface **30** via one or more fasteners, one or more adhesives, and combinations thereof. In another embodiment, the illumination PCB **32** may be set in a fixed position via the touch pad member **36** when secured to housing **12** as shown FIGS. **8**, **10** and **20**. In this embodiment, the illumination PCB **32** is disposed across the opening **35** in the floor **33** with the first push switch **50** and the second push switch **51** positioned apart at locations aligned with the push buttons **54** and **55** of the touch pad member **36** providing for operable contact between a pressed push button **54** or **55** and a corresponding push switch **50** or **51**.

[0116] In one embodiment, the touch pad member **36** may include an outer perimeter **37** having a size and shape corresponding to the mating surface **30** providing a seal there between when assembled. In one embodiment, the touch pad member **36** may be secured to the mating surface **30** via a snap-fit connection. In another embodiment, the touch pad member **36** may be secured to the mating surface **30** via one or more adhesives, e.g., one or more adhesives applied along the outer perimeter **37** of the touch pad member **36** for sealing the outer perimeter **37** to the mating surface **30**.

[0117] Referring to FIG. **12**, a power input sub-assembly of the enclosed reflex sight **11** may include a cylindrical battery compartment **38** operationally configured to hold a circular coin cell battery **43** as known in the art of optical sights. In one embodiment, the battery compartment **38** may be located on a second side **17** of the housing **12** as shown and comprise an assembly of battery compartment components as commonly included as part of a coin cell battery compartment **38**. In this embodiment, the second side **17** of the housing **12** includes a circular opening with a raised circular perimeter **39** extending out from the second side **17** of the housing **12** as a circular sidewall at a distance providing a cavity having inner dimensions suitable for housing battery compartment components and at least one coin cell battery **43** therein during operation. Because commercially available coin cell batteries known at the time of this disclosure are produced in various sizes, i.e., various diameters and heights, the raised circular perimeter **39** and battery compartment components housed therein may be manufactured to hold a coin cell battery **43** of predetermined dimensions for powering the enclosed reflex sight **11** according to voltage requirements of the control circuitry of the enclosed reflex sight **11**.

[0118] Exemplary battery compartment components housed within the raised circular perimeter **39** include at least a battery compartment PCB **45** with a negative terminal **46** for a coin cell battery **43** attachable to the battery compartment PCB **45**, a positive terminal **47** or “positive terminal ring **47**” for a coin cell battery **43**, a PCB retaining ring **48** for retaining the battery compartment PCB **45** and the positive terminal ring **47** within the battery compartment **38**, a removable threaded battery cover **42** comprising a removable polymeric seal **44**, e.g., a rubber O-ring, for sealing a coin cell battery **43** within the battery compartment **38** to protect the coin cell battery **43** against environmental influences such as dirt and moisture. In this embodiment, the battery compartment PCB **45**, the positive terminal ring **47**, the PCB retaining ring **48**, the removable seal **44** are each provided as circular type member with an outer diameter suitable for operation with an inner surface **41** of a raised circular perimeter **39** and a coin cell battery **43** of known dimensions. For example, as known in the art of coin cell battery compartments for optical sights, the negative terminal **46** (or “negative contact plate”) is operationally configured to contact an anode case of a coin cell battery **43** and the positive terminal ring **47** includes an inner surface operationally configured to contact a positive terminal such as a cathode case of a coin cell battery **43** to complete an electric circuit when the coin cell battery **43** is housed within the battery compartment **38**. A description of a coin cell battery for use with optical sights and a description of coin cell battery compartments of optical sights is provided in U.S. Pat. No. 11,236,971 B2, titled “Solar Powered Cap Assembly for Optical Sighting Systems,” issued on Feb. 1, 2022, which is herein incorporated by reference in its entirety.

[0119] In one non-limiting embodiment, the inner surface of the battery compartment **38** may comprise dimensions suitable for housing a CR2032 3.0 Volt (20.0 mm×3.2 mm) coin cell lithium battery as a power source of the enclosed reflex sight **11**. In another non-limiting embodiment, the inner surface of the battery compartment **38** may comprise dimensions suitable for housing a CR1632 3.0 Volt (16.0 mm×3.2 mm) coin cell lithium battery as a power source of the enclosed reflex sight **11**.

[0120] With further reference to FIG. **12**, components of the power input sub-assembly described above are assembled together within the battery compartment **38** by placing the assembled battery compartment PCB **45** and negative terminal **46** against a floor **38A** of the battery compartment **38** with the positive terminal ring **47** sandwiched between the battery compartment PCB **45** and the PCB retaining ring **48**. In one embodiment, the positive terminal ring **47** may include an outward annular flange **47A** operationally configured as an abutment surface for the PCB retaining ring **48**. In one embodiment, the PCB retaining ring **48** may include an outer threaded surface **48A** operationally configured to threadedly communicate with a threaded inner surface **41** of the raised perimeter **39** of the battery compartment **38** whereby the PCB retaining ring **48** may be threadedly communicated with the threaded inner surface **41** in a manner effective to form a close fit of the battery compartment PCB **45**, positive terminal ring **47**, and PCB retaining ring **48** within the battery compartment **38**. As further shown in FIG. **12**, a battery cover **42** may include a threaded surface **49** along the outer perimeter of the battery cover **42** for threadedly mating with a threaded inner surface **41** of the raised perimeter **39** of the battery compartment **38** in a manner effective to secure a coin cell battery **43** in the battery compartment **38** and maintain the electric circuit during operation of the enclosed reflex sight **11**. In another embodiment, the outer surface **40** of the raised perimeter **39** may include a threaded surface for threadedly mating with a threaded inner surface of a corresponding battery cover **42**.

[0121] In one embodiment, each of the negative terminal **46** and the positive terminal **47** for a coin cell battery **43** are in electrical communication with the battery compartment PCB **45** via direct contact and the battery compartment PCB **45** may be electrically communicated with the PCB **20** via electrical wiring or rigid flex connectors.

[0122] Suitably, the windage and elevation adjustment mechanisms are provided as an assembly of parts operationally configured for manual adjustment of the emitter assembly carrier **29** in both a vertical direction and a horizontal direction. As described below, by configuring the windage and elevation adjustment mechanisms to operate horizontally the windage and elevation adjustments are able to be located within the housing **12** at or near the same elevation as the mounting interface **14** located below a forward part of the enclosed reflex sight **11** as opposed to being located above a mounting interface as shown in the prior art described above. As such, an assembly **10** of this disclosure is operationally configured as a low-profile enclosed reflex sight **11** for co-witnessing with standard iron sights of a firearm to which the assembly **10** is mounted.

[0123] With reference to FIGS. **13** and **14**, a housing **12** of this disclosure generally comprises a box type inner surface including a ceiling or upper surface **60**, a floor or bottom surface **61**, a left sidewall or left side surface **62** (see FIG. **12**) and a right sidewall or right side surface **63**. In this embodiment, a forward part of the inner surface of the housing **12** comprises a front mating socket **56** for securing and aligning the front lens **16** to the inner surface of the housing **12** and a rearward part of the inner surface of the housing **12** comprises a rear mating socket **57** for securing and aligning the rear lens **18** to the inner surface of the housing **12**.

[0124] In one embodiment, one or more adhesives and/or fluid sealants may be included along the front mating socket **56** to form a seal along the perimeter **254** of the front lens **16** and along the rear mating socket **57** to form a seal with the perimeter **255** of the rear lens **18**. In another embodiment, one or more seals may be disposed between the front mating socket **56** and the perimeter **254** of the front lens **16** and the rear mating socket **57** and the perimeter **255** of the rear lens **18**.

[0125] With further reference to FIGS. **13** and **14**, the bottom of the housing **12** includes a first

body member **64** with an upper surface comprising a horizontal surface of the bottom surface **61** of the housing **12** including the part of the bottom surface **61** that comprises the bottom parts the front mating socket **56** and the rear mating socket **57**. The bottom of the housing **12** also includes a second body member **69** having an outer surface that forms part of the bottom outer surface **53** of the housing **12** and an inner surface **65** that forms part of the bottom surface **61** of the housing **12**. In this embodiment, the inner surface **65** comprises a funnel type sloping surface that extends downward from an innermost flange **58** of the front mating socket **56** to a transition **66** characterized by a vertical surface **67** extending down from the transition **66** forming part of a forward sidewall of a compartment **70** for holding the emitter assembly carrier **29** and the windage and elevation adjustment mechanisms (hereafter collectively referred to as the “adjustment assembly **100**”), which are operationally configured to move the emitter assembly **25** within the compartment **70** both vertically and horizontally.

[0126] Referring to FIG. **15**, the compartment **70** is provided as an open space with perimeter sidewalls **71-74** in a rectangular configuration and an upper surface or ceiling **75** disposed there between. In this embodiment, the compartment **70** comprises an inner surface having a plurality of engagement surfaces for receiving an adjustment assembly **100** of a particular size and shape in abutment thereto when the adjustment assembly **100** is sealed within the compartment via a bottom plate **76** fastened to the housing **12** (see FIG. **21**) wherein the bottom plate **76** acts as a floor of the compartment **70** during operation of the enclosed reflex sight **11**. In another embodiment, the configuration of the compartment **70** may vary to accommodate a different size and/or shape emitter assembly **25** and/or adjustment assembly **100**.

[0127] Referring to FIGS. **14** and **16**, in one embodiment an inner surface of a forward sidewall **74** of the compartment **70** may include forward engagement members **77** and **78** on opposing sides of the transition **66** and the vertical surface **67**. In this embodiment, the engagement members **77** and **78** extend down from the ceiling **75** from a point higher than the transition **66** whereby the engagement members **77** and **78**, the transition **66** and the ceiling **75** form an opening between the compartment **70** and the optical space **34** of the housing **12**. Suitably, an emitter assembly **25** is located in the compartment **70** at a point effective to project an operable reticle through the opening onto the inner surface **19** of the front lens **16** without encumbrance.

[0128] Referring to FIGS. **15-20**, the forward engagement members **77** and **78** include vertical forward engagement surfaces **79** and **80** for receiving front parts of the adjustment assembly **100** in abutment thereto. The compartment **70** includes one or more rear engagement members **81** comprising one or more vertical rear engagement surfaces **82-84** operationally configured to receive at least part of a rear part of the adjustment assembly **100** in abutment thereto. In this embodiment, the left sidewall **71** includes an inner surface **85** for receiving at least part of a left side part of the adjustment assembly **100** in abutment thereto and the right sidewall **73** includes an inner surface **86** for receiving at least part of a right side part of the adjustment assembly **100** in abutment thereto. The compartment **70** may also include one or more intermediate engagement members **87** and **88** extending out from the ceiling **75** having one or more vertical surfaces, e.g., see vertical surfaces **89** and **90** in FIG. **14**, operationally configured to receive one or more other parts of the adjustment assembly **100** in abutment thereto (see the embodiment of the adjustment assembly **100** housed within the compartment **70** in FIGS. **19** and **20**).

[0129] As shown in FIG. **20**, in one embodiment the forward engagement surfaces **79** and **80** and the one or more rear engagement members **81** (see FIG. **16**) may include threaded fastener holes **91-94** for receiving bottom plate fasteners **95** for releasably securing the bottom plate **76** to the compartment **70** (see through holes **96** of the bottom plate **76** in FIG. **30**).

[0130] As stated above, an adjustment assembly **100** of this disclosure includes a windage adjustment mechanism and an elevation adjustment mechanism both of which operate along a horizontal path effective to drive an emitter assembly carrier **29** of the adjustment assembly **100** both vertically up and down and horizontally left and right. The configuration of the windage

adjustment mechanism and the elevation adjustment mechanism allows the compartment **70** to be located at a point along the bottom rear of the housing **12** whereby the bottom surface **98** of the compartment **70**, i.e., the bottom surface of the perimeter sidewalls **71-74**, is the lowermost surface of the housing **12** and the lowermost surface of the enclosed reflex sight **11** (see plane **P1** in FIG. **22**; see also FIG. **23**). In other words, the configuration of the windage adjustment mechanism and elevation adjustment mechanism provides an adjustment assembly **100** that can be located in a compartment **70** comprising a bottom surface **98** forming a planar surface or a substantially planar surface with the bottom surface **99** of the mounting interface **14** when assembled (see plane **P2** in FIG. **24**). As FIGS. **24** and **25** illustrate, the mounting interface **14** does not extend the entire length of the enclosed reflex sight **11**. Rather, the entire adjustment assembly **100** is located within the compartment **70** at a position behind the mounting interface **14** wherein at least part of the adjustment assembly **100** is located below the uppermost surface **157** of the mounting interface **14**. As such, the bottom surface **200** of the assembly **10** is comprised of both the lowermost surface of the enclosed reflex sight **11**, i.e., bottom surface **98**, and the bottom surface **99** of the mounting interface **14** wherein the bottom surface **200** of the assembly **10** comprises a planar surface or a substantially planar surface operationally configured as an abutment surface with a bottom surface **570** of an upper mounting surface **557** of one or more firearms **550**. In one embodiment, the bottom surface **98** may be referred to as a rearward bottom surface of the enclosed reflex sight **11** and a bottom surface **179** of the housing **12** forward of the bottom surface **98** may be referred to as a forward bottom surface of the housing **12** (see FIG. **23**). In one embodiment, the bottom surface **179** may be referred to as a first bottom surface or upper bottom surface of the enclosed reflex sight **11** and the bottom surface **98** may be referred to as a second bottom surface or lowermost surface of the enclosed reflex sight **11**. In one embodiment, the bottom plate **76** may be referred to as part of the bottom surface **98**. In one embodiment, the plate member **22** may be referred to as part of the forward bottom surface **179** of the enclosed reflex sight **11**. In addition, a rear surface **300** of the enclosed reflex sight **11** may be referred to as the rear surface of the assembly **10** and a front surface **301** of the mounting interface **14** may be referred to as the front surface of the assembly **10**. Referring to FIGS. **57-60**, the front surface **301** may sit adjacent to or in abutment with at least part of the front wall surface **571** of an upper mounting surface **557** and the rear surface **300** may sit adjacent to or in abutment with at least part of the back wall surface **572** of an upper mounting surface **557** of a firearm **550**.

[0131] FIGS. **26-28** depict an embodiment of an adjustment assembly **100** operable with a housing **12** as described above. The adjustment assembly **100** includes an assembly of components including wedges, inclined plane surfaces or ramps, threaded members, and biasing members assembled in a manner whereby the adjustment assembly **100** is greater in width than in height suitable for operation within a compartment **70** as described above. As previously discussed above, an inner surface **97** of the bottom plate **76** acts as a floor of the compartment **70**. As such, the bottom plate **76** is operationally configured as a support surface for the adjustment assembly **100** and may be considered part of the adjustment assembly **100** for purposes of this disclosure.

[0132] Looking at FIG. **26**, the adjustment assembly **100** includes an elevation ramp member **102** and an emitter seat **104** having a bottom facing ramp surface **101** operationally configured to ride on a corresponding ramp surface **103** of the elevation ramp member **102**. As shown, the emitter seat **104** also includes a forward ledge type support surface **105** (“forward ledge **105**”) for holding an emitter assembly carrier **29**, which acts as a support for holding the emitter assembly **25**.

[0133] In addition to the emitter assembly carrier **29**, windage adjustment of the emitter assembly **25** is accomplished via windage adjustment components including (1) a threaded windage drive screw **110**, (2) a threaded windage adjustment key **112** for threadedly mating with the windage drive screw **110**, (3) a threaded windage retaining ring **114** for threadedly mating with the windage adjustment key **112**, (4) an O-ring seal **116** attachable to the windage adjustment key **112** and (5) a windage wedge member **118** operationally configured to bias the emitter assembly carrier **29** and

the windage drive screw **110** as described below.

[0134] In addition to the emitter assembly carrier **29**, the elevation ramp member **102** and the emitter seat **104**, elevation adjustment of the emitter assembly **25** is accomplished via elevation adjustment components including (1) a threaded elevation drive screw **120**, (2) a threaded elevation retaining ring **122**, (3) a threaded elevation adjustment key **124**, and (4) an O-ring seal **126** for the elevation adjustment key **124**.

[0135] In this embodiment, the emitter assembly carrier **29** includes a central support member **108** with opposing abutment members, e.g., a first abutment member **106** with a first wedge contact face **128** operationally configured to communicate with a wedge contact face **111** of a head **117** of the windage drive screw **110** and an opposing second abutment member **107** with a second wedge contact face **129** operationally configured to communicate with a wedge contact face **119** of the windage wedge member **118**. As shown, the central support member **108** includes a front planar face member **115** for receiving an emitter assembly **25** in abutment thereto via one or more adhesives. Suitably, the first abutment member **106** and the second abutment member **107** are spaced apart a distance effective to hold the forward ledge **105** in abutment there between preventing unwanted lateral movement of the emitter seat **104** when the emitter seat **104** is biased by the elevation ramp member **102** as described below. As such, in one embodiment the emitter assembly carrier **29** may be considered part of the windage adjustment mechanism of the adjustment assembly **100**.

[0136] Still referring to FIG. **26**, the adjustment assembly **100** comprises a plurality of biasing springs and corresponding plungers that apply tension to the inner surface of the compartment **70** to assist in maintaining the windage adjustment components and the elevation adjustment components in desired positions within the compartment **70** during operation of the adjustment assembly **100**. For example, the first body member **128** of the emitter assembly carrier **29** includes a vertical hole **130** for receiving a biasing spring **131** and a plunger **132** therein and the second body member **107** includes a vertical hole **134** for receiving a biasing spring **135** and a plunger **136** therein to bias the emitter assembly carrier **29** apart from the ceiling **75** of the compartment **70**. Likewise, the emitter seat **104** includes one or more vertical holes **138** and **139**, each vertical hole being operationally configured to receive a biasing spring and a plunger therein to bias the emitter seat **104** apart from the ceiling **75** (see biasing springs **140**, **141** and plungers **142**, **143**). In other words, when the adjustment assembly **100** is secured within the compartment **70** the emitter assembly carrier **29** and the emitter seat **104** are biased toward the bottom plate **76**. In one embodiment, the inner surfaces **85** and **86** and the ceiling **75** may include cavities (not shown) for holding the distal ends of the corresponding plungers.

[0137] In this embodiment, the elevation ramp member **102** includes a raised body member **145** comprising a horizontal hole **146** for receiving a biasing spring **147** and a plunger **148** therein to bias the elevation ramp member **102** apart from the inner surface **85** of the left sidewall **71**. The windage wedge member **118** also includes one or more horizontal holes **150** and **151** (see FIG. **29**) for receiving a biasing spring and plunger therein to bias the windage wedge member **118** apart from the inner surface **86** of the right sidewall **73** (see biasing springs **152**, **153** and plungers **154**, **155**).

[0138] Referring again to FIGS. **17** and **18**, the adjustment assembly **100** is further secured to the compartment **70** by way of the windage adjustment key **112** and the elevation adjustment key **124** which are each communicated with apertures in the housing **12**. In particular, the left sidewall **71** includes an aperture **158** with an inner cylindrical perimeter surface for mating with a circular head **113** of the windage adjustment key **112** and the right sidewall **73** includes an aperture **159** with an inner cylindrical perimeter surface for mating with a circular head **125** of the elevation adjustment key **124**. In particular, the windage adjustment key **112** includes a round detent **161** and a corresponding biasing spring **162** disposed radially in a hole **165** of the circular head **113** and the elevation adjustment key **124** includes a round detent **166** and a corresponding biasing spring **167**

disposed radially in a hole **168** of the circular head **125** for mating with raised splines disposed 360.0 degrees along the inner perimeter surface of each of the apertures **158** and **159** (see the raised splines **160** and **170** in FIGS. **17** and **18**). In operation, the detents **161** and **166** operate as ball plungers that are biased between the raised teeth **160** and **170** under spring tension necessary to hold the windage adjustment key **112** and the elevation adjustment key **124** in a fixed position until manually turned to disengage the detents **161** and **166** when adjusting the windage and/or elevation. In addition, the configuration of the detents **161**, **166** and the raised teeth **160**, **170** produces a clicking sound and/or a clicking feel to a user when turning the windage adjustment key **112** and the elevation adjustment key **124**.

[0139] In one embodiment, the circular heads **113** and **125** of the windage adjustment key **112** and the elevation adjustment key **124** may each comprise a drive **172**, **173** to assist in manually turning the adjustment keys **112** and **124**, e.g., using a slotted flat blade screw driver or an adjustment tool as known in the art of optical sights. In addition, a shank of each of the adjustment keys **112** and **124** may include a concentric groove adjacent the circular heads **112** and **125** for holding an O-ring seal **116**, **126** to form a seal along the inner surfaces **177**, **178** of the apertures **158** and **159** to protect against environmental influences such as dirt and moisture from entering the housing **12** through the apertures **158** and **159** (see grooves **175** and **176** in FIG. **26**).

[0140] In one embodiment, the windage retaining ring **114** is permanently fixed to an outer threaded surface **156** of the adjustment key **112** and the elevation retaining ring **122** is permanently fixed to an outer threaded surface **137** of the adjustment key **124** whereby the windage retaining ring **114** and the elevation retaining ring **122** are operationally configured to further secure the adjustment assembly **100** to the housing **12** by similarly acting as stop type members preventing the windage adjustment key **112** and the elevation adjustment key **124** from exiting out from the apertures **158**, **159**. For example, referring to the elevation adjustment key **124** as shown in FIG. **27**, the inner surface **178** of the aperture **159** includes a concentric flange **169** operationally configured as an abutment surface for the elevation retaining ring **122** whereby the concentric flange **169** and the elevation retaining ring **122** operate collectively as a stop for catching the head **174** of the elevation drive screw **120** preventing the elevation adjustment key **124** from being directed out from the housing **12** through the aperture **159**. The elevation retaining ring **122** and the flange **169** are also operationally configured to maintain the outer face **133** of the circular head **125** of the elevation adjustment key **124** flush with the adjacent outer surface of the housing **12** (see FIG. **21**). A similar flange in the inner surface **177** of aperture **158** and the windage retaining ring **114** operate as described above to prevent the head **117** of the windage drive screw **110** from being directed out from the housing **12** through the aperture **158**.

[0141] To minimize the overall size of the enclosed reflex sight **11**, the housing **12** materials including the bottom plate **76** and adjustment assembly **100** components are suitably configured in as small a package allowable for desired functioning of the enclosed reflex sight **11** without realizing structural failure of one or more of the housing **12**, bottom plate **76** and adjustment assembly **100** components. As an example, the perimeter of the bottom plate **76** and the perimeter of the opening of the compartment **70** may include corresponding indexing features **181**, **182** and **183**, **184** (see FIGS. **30** and **31**) in order to provide added material thickness to the bottom plate **76** and the bottom surface **98** of the compartment **70** at an abutment **185** between the bottom plate **76** and the bottom surface **98** for providing structural strength to the lower most part of the flange in each of the apertures **158** and **159** (see FIG. **27**).

[0142] To adjust the emitter assembly **25** vertically up and down, the elevation adjustment key **124**, which comprises a shaft having (1) an inner threaded surface **127** in threaded communication with an outer threaded surface **121** of a shaft of the elevation drive screw **120** and (2) an outer threaded surface **137** in threaded communication with an inner threaded surface **144** of the elevation retaining ring **122** may be turned clockwise and counter-clockwise thereby acting on the outer threaded surface **121** to direct the elevation drive screw **120** in a linear direction. For example, to

direct the emitter assembly **25** linearly in an upward direction (see directional arrow **A1** in FIG. **28**), the elevation adjustment key **124** may be turned counter-clockwise directing the head **174** of the elevation drive screw **120** linearly toward the inner surface **85** of the left sidewall **71** against a right side vertical surface **109** of the elevation ramp member **102** (see directional arrow **A2** in FIG. **28**) causing the ramp surface **103** of the elevation ramp member **102** to act on the ramp surface **101** of the emitter seat **104** pushing the emitter seat **104** upward (see directional arrow **A1**) via a sliding coupling between the ramp surface **101** and the ramp surface **103**. As the emitter seat **104** is directed in an upward direction, the emitter assembly carrier **29** and the emitter assembly **25** secured thereto are simultaneously directed in an upward direction a distance equal to the distance traveled by the emitter seat **104**. Suitably, the detent **166** and biasing spring **167** are operationally configured to engage the raised splines **170** to assist in fixing the location of the emitter assembly **25**, biasing springs **152**, **153** and plungers **154**, **155** during use of the enclosed reflex sight **11**.

[0143] To direct the emitter assembly **25** in a downward direction (see directional arrow **A3** in FIG. **28**), the elevation adjustment key **124** may be turned clockwise as desired to direct the elevation drive screw **120** linearly toward the inner surface **86** of the right sidewall **73** whereby the elevation ramp member **102** is simultaneously directed linearly toward the inner surface **86** of the right sidewall **73** via a biasing force of the biasing spring **147** mated to the elevation drive screw **120** allowing the emitter seat **104** to slide down the ramp surface **103** to a desired elevation of the emitter assembly **25** along the ramp surface **103**. Accordingly, rotational movement of the elevation adjustment key **124** is converted into linear movement of the emitter assembly **25**.

[0144] To adjust the emitter assembly **25** horizontally left and right, the windage adjustment key **112**, which comprises a shaft having (1) an inner threaded surface (not shown) similar as inner threaded surface **127** in threaded communication with an outer threaded surface **149** of a shaft of the windage drive screw **110** and (2) an outer threaded surface **156** in threaded communication with an inner threaded surface **171** of the windage retaining ring **114** may be turned clockwise and counter-clockwise thereby acting on the outer threaded surface **149** to direct the windage drive screw **110** in a linear direction. For example, to direct the emitter assembly **25** linearly toward the inner surface **86** of the right sidewall **73** (see directional arrow **A5** in FIG. **29**), the windage adjustment key **112** may be turned counter-clockwise directing the wedge contact face **111** of the head **117** of the windage drive screw **110** linearly against the first wedge contact face **128** of the first abutment member **106** of the emitter assembly carrier **29** to direct the emitter assembly carrier **29** toward the inner surface **86** of the right sidewall **73** according to (1) a sliding coupling between the wedge contact face **111** of the head **117** of the windage drive screw **110** and the first wedge contact face **128** of the first abutment member **106** and (2) a sliding coupling between the second wedge face **129** of the second abutment member **107** and the wedge face **119** of the windage wedge member **118**, which further directs each of the emitter assembly carrier **29** and the windage wedge member **118** linearly toward the inner surface **86** of the right sidewall **73**.

[0145] To direct the emitter assembly **25** toward the inner surface **85** of the left sidewall **71** (see directional arrow **A6** in FIG. **29**), the windage adjustment key **112** may be turned clockwise as desired to direct the windage drive screw **110** linearly toward the inner surface **85** of the left sidewall **71**, whereby a biasing force of the biasing springs **152**, **153** of the windage wedge member **118** directs the wedge face **119** of the windage wedge member **118** against the second wedge face **129** of the second abutment member **107**, which further directs the first wedge face **128** of the first abutment member **106** against the wedge face **111** of the head **117** of the windage drive screw **110** in a sliding action until the emitter assembly **25** realizes a desired position within the compartment **70**. Accordingly, rotational movement of the windage adjustment key **112** is converted into linear movement of the emitter assembly **25**.

[0146] As stated above a mounting interface **14** of the assembly **10** is positioned forward of the adjustment assembly **100** housed in the compartment **70** when the mounting interface **14** is assembled to the enclosed reflex sight **11**. As such, the attachment points on the enclosed reflex

sight **11** for receiving an amounting interface **14** of this disclosure are also positioned forward of the adjustment assembly **100** of the enclosed reflex sight **11**. As shown in FIGS. **11** and **15**, the housing **12** includes two forward attachment points for a mounting interface **14** provided as fastener holes **187** and **188** disposed vertically through the first body member **64** of the housing **12** forward of the front lens **16**. The housing **12** further includes a left side attachment point for a mounting interface **14** provided as a horizontal threaded fastener hole **189** located forward the adjustment assembly **100** at a point higher along the housing **12** than the bottom surface **98** (see FIG. **22**) and a right side attachment point for a mounting interface **14** provided as a horizontal threaded fastener hole **190** located forward the adjustment assembly **100** at a point higher along the housing **12** than the bottom surface **98** (see FIG. **23**). As further shown in FIGS. **22** and **23**, the fastener holes **187** and **188** are also located at a point higher along the housing **12** than the bottom surface **98**.

[0147] As stated above, an enclosed reflex sight **11** of this disclosure may be used as part of an assembly **10** with one or more mounting interfaces **14** operationally configured for use with a plurality of upper mounting surface footprints as known in the art of firearms, including one or more upper mounting surface **557** footprints on semi-automatic pistol slides, for example, one or more machined, drilled or milled surfaces of semi-automatic pistol slides. Non-limiting examples of upper mounting surface **557** footprints that the one or more mounting interfaces **14** may be configured for use with include one or more dovetail footprints, the Glock® Modular Optic System plate footprint (or “Glock® MOS” plate footprint); Glock, Inc., Smyrna, Georgia, U.S.A. and the Trijicon® RMR® Reflex Sight footprint, Trijicon, Inc. Wixom, Michigan, USA.

[0148] FIGS. **32-34** illustrate a first non-limiting embodiment of a mounting interface **14** of this disclosure operationally configured for use with a Glock® MOS plate footprint. FIGS. **8-10** depict the embodiment of the mounting interface **14** of FIGS. **32-34** assembled to an enclosed reflex sight **11** of this disclosure.

[0149] FIGS. **35-37** illustrate a second non-limiting embodiment of a mounting interface **14** of this disclosure operationally configured for use with an embodiment of a Trijicon® RMR® Reflex Sight footprint. FIGS. **38-40** illustrate a third non-limiting embodiment of a mounting interface of this disclosure operationally configured for use with another embodiment of a Trijicon® RMR® Reflex Sight footprint.

[0150] Common to each embodiment of mounting interface **14** is an upper surface **192** and a bottom surface **99**. Suitably, at least part of the upper surface **192** is operationally configured as an abutment surface for at least part of the enclosed reflex sight **11**, e.g., at least part of the bottom surface **179** of the enclosed reflex sight **11**. Likewise, at least part of the bottom surface **99** is operationally configured as an abutment surface for at least part of a bottom surface **570** of an upper mounting surface **557** of a firearm **550**.

[0151] Also common to each embodiment of mounting interface **14** is a fastener hole pattern corresponding to the pattern of fastener holes **187**, **188**, **189**, **190** of the enclosed reflex sight **11**. For example, each embodiment of the mounting interface **14** includes two forward vertical threaded fastener holes **194** and **195** corresponding to fastener holes **187** and **188**, a left side horizontal fastener hole **196** corresponding to threaded fastener hole **189** and a right side horizontal fastener hole **197** corresponding to threaded fastener hole **190**. As shown, fastener hole **196** is disposed through a raised left sidewall **201** of the mounting interface **14** and fastener hole **197** is disposed through a raised right sidewall **202** of the mounting interface **14**. In addition to providing substrates for providing fastener holes **196** and **197** the left sidewall **201** and the right sidewall **202** are operationally configured to cradle a corresponding enclosed reflex sight **11** when assembled to assist in maintaining the enclosed reflex sight **11** in a fixed position during operation. Fastener holes **194** and **195** may be referred to herein as front mounts and fastener holes **196** and **197** may be referred to as side mounts for fasteners, e.g., threaded screws.

[0152] As further depicted, each embodiment of mounting interface **14** may include one or more through holes corresponding to a mounting standard, e.g., a fastener hole pattern and socket

pattern, of a particular firearm mounting footprint for receiving corresponding fasteners **186**. For example, the embodiment of FIGS. **32-34** includes two opposing countersunk apertures **205** and **206** disposed at the center or approximate center of the mounting interface **14** corresponding to fastener holes of a firearm. The embodiment of FIGS. **35-37** includes two opposing countersunk apertures **207** and **208** disposed at a rearward part of the mounting interface **14**. The embodiment of FIGS. **38-40** includes two opposing raised countersunk apertures **209** and **210** disposed at a rearward part of the mounting interface **14**. In one embodiment, the fastener holes of a mounting interface **14** may also be referred to as bolt holes or screw holes.

[0153] A housing **12**, mounting interface **14** and components comprising the adjustment assembly **100** may be constructed of one or more materials durable for one or more operations and/or as may be required by law or regulation. Suitable materials of construction may include, but are not necessarily limited to those materials resistant to chipping, cracking, excessive bending and reshaping as a result of ozone, weathering, heat, moisture, other outside mechanical and chemical influences, as well as physical impacts. Suitably materials of construction may include one or more metals, plastics, rubbers, woods, filled composite materials, and combinations thereof. Suitable metals include, but are not necessarily limited to stainless steel, hardened steel, mild steel, aluminum, copper, nickel, brass, titanium, and combinations thereof. Suitable plastics include, but are not necessarily limited to glass-filled polymers, durable plastic composite materials, and combinations thereof. One suitable glass-filled polymer includes, but is not necessarily limited to glass-filled nylon. One or more adhesives and/or sealants may be used to secure and/or seal one or more components described herein. In addition, one or more heat treatments and/or one or more coatings may be applied to the housing **12**, the mounting interface **14** and one or more other components described herein.

[0154] The disclosure will be better understood with reference to the following non-limiting examples, which are illustrative only and not intended to limit the present disclosure to a particular embodiment.

Example 1

[0155] In a first non-limiting example, an embodiment of an enclosed reflex sight **11** is provided as shown in FIGS. **41-48**. The outer surface of the housing **12** is not limited to the ornamental configuration as shown, but may vary in another embodiment of the housing **12**.

Example 2

[0156] In a second non-limiting example, an assembly **10** as shown in FIGS. **49-56** is mounted within an upper mounting surface **557** of a pistol slide **575** (see FIGS. **57-60**) and the altitude or elevation **580** of a lowermost part **560** of the rear lens **18** is compared to the altitude or elevation **581** of an uppermost surface of a standard height rear sight **577** of the pistol slide **575**. As shown in FIG. **61**, the altitude or elevation of the lowermost part **560** of the rear lens **18** is lower than the altitude or elevation **581** of the uppermost surface of the standard height rear sight **577**.

[0157] The assembly **10** is removed from the pistol slide **575** and replaced with an enclosed reflex sight **500** as described in reference to FIGS. **3-5**. The altitude or elevation **582** of the lowermost part **515** of the rear lens **516** of the enclosed reflex sight **500** is compared to the altitude or elevation **581** of the uppermost surface of the standard height rear sight **577** of the pistol slide **575**. As shown in FIG. **61**, the altitude or elevation of the lowermost part **515** of the rear lens **516** is higher than the altitude or elevation **581** of the uppermost surface of the standard height rear sight **577**.

Example 3

[0158] With reference to FIGS. **62** and **63**, in a third non-limiting example an enclosed reflex sight **11** of an assembly **10** as shown in FIGS. **58-60** may comprise the dimensions as shown in Table 1. TABLE-US-00001 TABLE 1 D1: 6.44 mm (0.25 inches); D2: 31.63 mm (1.25 inches); D3: 26.42 mm (1.04 inches); D4: 45.21 mm (1.78 inches); and D5: 2.15 mm (0.08 inches).

[0159] An assembly **10** and system of the present disclosure may be described according to one or

more of the following Embodiments.

[0160] Embodiment 1. An assembly, comprising: [0161] an enclosed reflex sight; and [0162] a mounting interface; [0163] wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight; [0164] wherein the enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the mounting interface; and [0165] wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.

[0166] Embodiment 2. The assembly of Embodiment 1, wherein the assembly includes a first maximum height and wherein the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0167] Embodiment 3. The assembly of Embodiment 1, wherein the assembly includes a maximum height and wherein the height of the enclosed reflex sight from an uppermost surface of the enclosed reflex sight to the lowermost surface of the enclosed reflex sight is the same or substantially similar as the maximum height of the assembly.

[0168] Embodiment 4. The assembly of Embodiment 1, wherein a first distance between an optical center of a clear aperture of the enclosed reflex sight and the lowermost surface of the enclosed reflex sight is the same or substantially similar as a second distance between the optical center of the clear aperture and a bottommost surface of the mounting interface.

[0169] Embodiment 5. An assembly, comprising: [0170] an enclosed reflex sight; and [0171] a mounting interface; [0172] wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the enclosed reflex sight is located at a position closer to the upper mounting surface of the firearm than at least part of the mounting interface and the assembly is located forward of a rear end surface of the firearm.

[0173] Embodiment 6. The assembly of Embodiment 5, wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0174] Embodiment 7. The assembly of Embodiment 5, wherein a distance between an optical center of a clear aperture of the enclosed reflex sight and the lowermost surface of the enclosed reflex sight is the same or substantially similar as a distance between the optical center of the clear aperture and a bottommost part of the mounting interface.

[0175] Embodiment 8. The assembly of Embodiment 5, wherein the enclosed reflex sight comprises a first length and the mounting interface includes a second length less than the first length.

[0176] Embodiment 9. The assembly of Embodiment 5, wherein the enclosed reflex sight includes a windage adjustment mechanism and an elevation adjustment mechanism that are located behind the mounting interface.

[0177] Embodiment 10. An assembly, comprising: [0178] an enclosed reflex sight; and [0179] a plurality of mounting interfaces; [0180] wherein each of the plurality of mounting interfaces is configured to secure the assembly to one or more of a plurality of firearms comprising a plurality of upper mounting surface footprints; [0181] wherein when the assembly is secured to an upper mounting surface of a first firearm of the plurality of firearms then the assembly is located on top of the first firearm forward of a rear end surface of the first firearm; and [0182] wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0183] Embodiment 11. The assembly of Embodiment 10, wherein when the assembly is secured to the upper mounting surface of the first firearm then the mounting interface of the assembly is located forward of at least part of the enclosed reflex sight.

[0184] Embodiment 12. The assembly of Embodiment 10, wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight.

[0185] Embodiment 13. An assembly, comprising: [0186] an enclosed reflex sight; and [0187] a plurality of mounting interfaces, each of the plurality of mounting interfaces being configured to secure the enclosed reflex sight to one or more firearms of a plurality of firearms comprising a plurality of upper mounting surface footprints; [0188] wherein when the assembly is secured to an upper mounting surface of a first firearm of the plurality of firearms then the assembly is located on top of the first firearm forward of a rear end surface of the first firearm; and [0189] wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0190] Embodiment 14. An assembly, comprising: [0191] an enclosed reflex sight; and [0192] a mounting interface for securing the enclosed reflex sight to an upper mounting surface of a firearm; [0193] wherein the enclosed reflex sight includes a first bottom surface at a first altitude or elevation and a second bottom surface at a second altitude or elevation lower than the first altitude or elevation; and [0194] wherein when the assembly of the enclosed reflex sight and the mounting interface is in a secured position with an upper mounting surface of a firearm then the first bottom surface and the second bottom surface of the enclosed reflex sight are located forward of a rear end surface of the firearm.

[0195] Embodiment 15. The assembly of Embodiment 14, wherein the mounting interface includes an upper surface operationally configured as an abutment surface for the first bottom surface of the enclosed reflex sight.

[0196] Embodiment 16. An assembly, comprising: [0197] an enclosed reflex sight; and [0198] a mounting interface for securing the enclosed reflex sight to an upper mounting surface of a firearm; [0199] wherein the enclosed reflex sight includes a first bottom surface at a first altitude or elevation and a second bottom surface at a second altitude or elevation lower than the first altitude or elevation; and [0200] wherein when the assembly is secured to an upper mounting surface of a firearm comprising a protruding rear sight then the first bottom surface and the second bottom surface of the enclosed reflex sight are located forward of protruding rear sight.

[0201] Embodiment 17. The assembly of Embodiment 16, wherein the mounting interface includes an upper surface operationally configured as an abutment surface for the first bottom surface of the enclosed reflex sight.

[0202] Embodiment 18. The assembly of Embodiment 16, wherein the mounting interface includes a bottom surface operationally configured as an abutment surface for an upper mounting surface of the firearm and wherein when the assembly is secured to the upper mounting surface of the firearm then the second bottom surface of the enclosed reflex sight is in planar alignment or substantially planar alignment with the bottom surface of the mounting interface.

[0203] Embodiment 19. An assembly, comprising: [0204] an enclosed reflex sight; and [0205] a mounting interface; [0206] wherein the enclosed reflex sight includes a lowermost surface; [0207] wherein when the assembly is secured to an upper mounting surface of a firearm then the entire assembly is located on top of the firearm; and [0208] wherein when the firearm is oriented in a firing position then the lowermost surface of the enclosed reflex sight is located at an altitude or elevation lower than at least part of the mounting interface.

[0209] Embodiment 20. The assembly of Embodiment 19, wherein the assembly includes a maximum height and the enclosed reflex sight includes a maximum height the same or substantially similar as the maximum height of the assembly.

[0210] Embodiment 21. The assembly of Embodiment 19, wherein the enclosed reflex sight includes a clear aperture with an optical center and wherein a distance between the optical center of the clear aperture of the enclosed reflex sight and the lowermost surface of the enclosed reflex sight is the same or substantially similar as a distance between the optical center of the clear aperture and

a bottommost part of the mounting interface.

[0211] Embodiment 22. An assembly, comprising: [0212] an enclosed reflex sight; and [0213] a mounting interface; [0214] wherein the enclosed reflex sight includes a lowermost surface; [0215] wherein when the assembly is secured to an upper mounting surface of a firearm and when the firearm is oriented in a firing position then the lowermost surface of the enclosed reflex sight is located at an altitude or elevation lower than at least part of the mounting interface; and [0216] wherein when the assembly is secured to the upper mounting surface of a firearm comprising a protruding rear sight then the entire assembly is located forward of the protruding rear sight of the firearm.

[0217] Embodiment 23. An assembly, comprising: [0218] an enclosed reflex sight; and [0219] a mounting interface; [0220] wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the enclosed reflex sight is located at a distance from the upper mounting surface of the firearm less than a distance of at least part of the mounting interface from the upper mounting surface and the assembly is located forward of a rear end surface of the firearm.

[0221] Embodiment 24. An assembly, comprising: [0222] an enclosed reflex sight; and [0223] a mounting interface; [0224] wherein the enclosed reflex sight includes a lowermost surface; and [0225] wherein when the assembly is secured to an upper mounting surface of a firearm oriented in a firing position then the lowermost surface of the enclosed reflex sight is located at an altitude lower than at least part of the mounting interface and the assembly is located forward of a rear end surface of the firearm.

[0226] Embodiment 25. A system, comprising: [0227] at least one firearm; and [0228] an assembly comprising: [0229] an enclosed reflex sight; and [0230] at least one mounting interface; [0231] wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight; [0232] wherein the enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the at least one mounting interface; and [0233] wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.

[0234] Embodiment 26. The system of Embodiment 25, wherein the assembly includes a first maximum height and wherein the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0235] Embodiment 27. The system of Embodiment 25, wherein a first distance between an optical center of a clear aperture of the enclosed reflex sight and the bottom surface of the rearward bottom portion of the enclosed reflex sight is the same or substantially similar as a second distance between the optical center of the clear aperture and a bottommost surface of the at least one mounting interface.

[0236] Embodiment 28. The system of Embodiment 25, wherein the system comprises two or more mounting interfaces.

[0237] Embodiment 29. A system comprising: [0238] at least one firearm comprising an upper mounting surface; and [0239] an assembly securable to the upper mounting surface comprising: [0240] an enclosed reflex sight; and [0241] at least one firearm mounting interface; [0242] wherein the assembly includes a maximum height and wherein the enclosed reflex sight includes a maximum height the same or substantially similar as the maximum height of the assembly; and [0243] wherein the assembly is located forward of a rear end surface of the firearm.

[0244] Embodiment 30. The system of Embodiment 29, the system comprising two or more firearm mounting interfaces.

[0245] Embodiment 31. The system of Embodiment 30, wherein each firearm mounting interface

of the two or more firearm mounting interfaces is configured for use with a different upper mounting surface footprint.

[0246] Embodiment 32. The system of Embodiment 29, wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height.

[0247] Embodiment 33. The system of Embodiment 29, wherein a first distance between an optical center of a clear aperture of the enclosed reflex sight and a lowermost surface of the enclosed reflex sight is the same or substantially similar as a second distance between the optical center of the clear aperture and a bottommost part of the at least one firearm mounting interface.

[0248] Embodiment 34. The system of Embodiment 29, wherein the enclosed reflex sight comprises a first length and the mounting interface includes a second length less than the first length.

[0249] Embodiment 35. The system of Embodiment 29, wherein the enclosed reflex sight includes a windage adjustment mechanism and an elevation adjustment mechanism that are located behind the at least one firearm mounting interface.

[0250] Embodiment 36. The system of Embodiment 29, wherein when the assembly is secured to the at least one firearm then the at least one firearm mounting interface is located forward of at least part of the enclosed reflex sight.

[0251] Embodiment 37. A system, comprising: [0252] a plurality of firearms providing at least two different upper mounting surfaces; [0253] an enclosed reflex sight; and [0254] a plurality of mounting interfaces, each mounting interface of the plurality of mounting interfaces being configured to secure the enclosed reflex sight to at least one upper mounting surface of the at least two different upper mounting surfaces; [0255] wherein when the enclosed reflex sight is secured to the at least one upper mounting surface of a first firearm of the plurality of firearms via a first mounting interface of the plurality of mounting interfaces then the enclosed reflex sight is located on top of the first firearm forward of a rear surface of the first firearm wherein a lowermost surface of the enclosed reflex sight is located closer to the at least one upper mounting surface than at least part of the first mounting interface.

[0256] Embodiment 38. The system of Embodiment 35, wherein the enclosed reflex sight includes a rearward bottom portion that houses an adjustment assembly of the enclosed reflex sight, the rearward bottom portion comprising the lowermost surface of the enclosed reflex sight.

[0257] Embodiment 39. A system, comprising: [0258] at least one firearm; and [0259] an assembly comprising: [0260] an enclosed reflex sight; and [0261] at least one mounting interface; [0262] wherein the enclosed reflex sight comprises windage and elevation adjustment mechanisms located in a rearward bottom portion of the enclosed reflex sight; [0263] wherein when the assembly is mounted to the at least one firearm oriented in a firing position then a bottom surface of the rearward bottom portion of the enclosed reflex sight is located at an altitude lower than at least part of the mounting interface; and [0264] wherein the rearward bottom portion of the enclosed reflex sight is located forward of a rear end surface of the firearm.

[0265] Embodiment 40. A system, comprising: [0266] a plurality of firearms providing at least two different upper mounting surfaces; one or more enclosed reflex sights; and [0267] a plurality of mounting interfaces, each mounting interface of the plurality of mounting interfaces being operationally configured to secure at least one of the one or more enclosed reflex sights to at least one upper mounting surface of one or more firearms of the plurality of firearms; [0268] wherein when a first enclosed reflex sight of the one or more enclosed reflex sights is secured to a first firearm of the plurality of firearms via a first mounting interface of the plurality of mounting interfaces and when the first firearm is oriented in a firing position then the enclosed reflex sight includes a lowermost surface that is located at an altitude or elevation lower than at least part of the first mounting interface; and the first enclosed reflex sight is located on top of the first firearm forward of a rear end surface of the first firearm.

[0269] Embodiment 41. A method of aiming a firearm at a target, comprising presenting at the target the firearm comprising an enclosed reflex sight and firearm mounting interface assembly wherein the maximum height of the enclosed reflex sight is the same or substantially similar as the maximum height of the assembly of the enclosed reflex sight and the firearm mounting interface.

[0270] Although the present disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead might be applied, alone or in various combinations, to one or more other embodiments whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the claimed invention should not be limited by any of the above-described embodiments.

[0271] Terms and phrases used in this disclosure, and variations thereof, unless otherwise expressly stated, should be construed as open-ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as meaning “including, without limitation” or the like, the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof, the terms “a” or “an” should be read as meaning “at least one,” “one or more,” “one or a plurality” or the like.

[0272] Persons of ordinary skill in the art will recognize that many modifications may be made to the present disclosure without departing from the spirit and scope of the disclosure. The embodiment(s) described herein are meant to be illustrative only and should not be taken as limiting the invention, which is defined in the claims.

Claims

1. An assembly, comprising: an enclosed reflex sight; and a mounting interface; wherein the enclosed reflex sight includes a rearward bottom portion located behind the mounting interface that houses an adjustment assembly of the enclosed reflex sight and wherein the rearward bottom portion defines a lowermost surface of the enclosed reflex sight; wherein the enclosed reflex sight includes an upper bottom surface forward of the rearward bottom portion of the enclosed reflex sight, the upper bottom surface comprising an abutment surface for the mounting interface; and wherein when the assembly is secured to an upper mounting surface of a firearm then the lowermost surface of the rearward bottom portion of the enclosed reflex sight is closer to the upper mounting surface of the firearm than the upper bottom surface of the enclosed reflex sight.
2. The assembly of claim 1, wherein the enclosed reflex sight includes a rear surface defining a rear surface of the assembly and wherein the mounting interface includes a front surface defining a front surface of the assembly.
3. The assembly of claim 1, wherein the enclosed reflex sight includes a front lens and a rear lens, wherein a front surface of the mounting interface is located forward of the front lens.
4. The assembly of claim 1, wherein the adjustment assembly includes a windage adjustment mechanism and an elevation adjustment mechanism located behind the mounting interface.
5. The assembly of claim 1, wherein the assembly has a first maximum height and wherein the enclosed reflex sight has a second maximum height the same as the first maximum height.
6. The assembly of claim 1, wherein the assembly includes a first maximum height and wherein the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height within $\pm 5\%$ of the first maximum height.
7. The assembly of claim 1, wherein the mounting interface has a bottom surface, wherein at least part of the bottom surface is operationally configured as an abutment surface for at least part of the upper mounting surface of the firearm and wherein when the assembly is secured to the upper mounting surface of the firearm then the second bottom surface of the enclosed reflex sight is in

planar alignment with the bottom surface of the mounting interface.

8. The assembly of claim 7, wherein when the assembly is secured to the upper mounting surface of the firearm then at least part of the second bottom surface of the enclosed reflex sight and at least part of the bottom surface of the mounting interface are in abutment with the upper mounting surface of the firearm.

9. An assembly, comprising: an enclosed reflex sight; and a mounting interface; wherein the enclosed reflex sight includes a rear surface defining a rear surface of the assembly and wherein the mounting interface includes a front surface defining a front surface of the assembly; and wherein when the assembly is secured to an upper mounting surface of a firearm then a lowermost surface of the enclosed reflex sight is located behind the mounting interface at a position closer to the upper mounting surface of the firearm than at least part of the mounting interface and wherein the assembly is located forward of a rear end surface of the firearm.

10. The assembly of claim 9, wherein the enclosed reflex sight includes a front lens and a rear lens, wherein the front surface of the mounting interface is located forward of the front lens.

11. The assembly of claim 9, wherein the assembly includes a first maximum height and the enclosed reflex sight includes a second maximum height the same or substantially similar as the first maximum height within $\pm 5\%$ of the first maximum height.

12. The assembly of claim 9, wherein the assembly comprises a first length, the enclosed reflex sight comprises a second length and the mounting interface comprises a third length, wherein the second length and the third length are each less than the first length.
